

Name: _____



Starter quiz

The intercept point

1 Which of these sequences are linear? Tick **3** correct answers

☐ $3n + 1$

☐ $7 - 5n$

☐ n^2

☐ $5n^3 - 15$

☐ $2n - 5$

2 Which of these best describes an intercept point? Tick **1** correct answer

☐ A point on the line where the x and y coordinates are the same

☐ A point on the line which you use to find the gradient.

☐ A point on the line where both values are positive.

☐ A point on the line where it meets an axis.



3 The tables shown have been labeled A,B,C,D,E. Match the table with the correct gradient. Write the correct letter in each box

A	<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>4</td><td>6</td><td>8</td></tr></table>	x	-1	0	1	y	4	6	8
x	-1	0	1						
y	4	6	8						
B	<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>-7</td><td>-9</td><td>-11</td></tr></table>	x	-1	0	1	y	-7	-9	-11
x	-1	0	1						
y	-7	-9	-11						
C	<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>12</td><td>8</td><td>4</td></tr></table>	x	-1	0	1	y	12	8	4
x	-1	0	1						
y	12	8	4						
D	<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>-11</td><td>-6</td><td>-1</td></tr></table>	x	-1	0	1	y	-11	-6	-1
x	-1	0	1						
y	-11	-6	-1						
E	<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y</td><td>-10</td><td>-6</td><td>-2</td></tr></table>	x	-1	0	1	y	-10	-6	-2
x	-1	0	1						
y	-10	-6	-2						

a	A
b	B
c	C
d	D
e	E

	4
	5
	-2
	-4
	2

4 In a linear sequence the common difference is 3 and the tenth term is 32. What is the first term? Fill in the blank

Term number	1	2	3	4	5	6	7	8	9	10
Term	?									32


+ 3

- 5 What is the gradient of the line which passes through the coordinates (4, 7) and (6,11) ?
Fill in the blank

- 6 What is the gradient of the line which passes through the coordinates (2, -6) and (0,-1) ?
Tick **1** correct answer

☐ -3

☐ $-\frac{5}{2}$

☐ $-\frac{2}{5}$

☐ $\frac{2}{5}$

☐ $\frac{5}{2}$

